

Alvyn Kwon

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EDUCATION

Georgia Institute of Technology

August 2023 - May 2028

Bachelor's and Master's of Science (B.S./M.S.) in Computer Science, GPA: 3.80/4.00

Atlanta, Georgia

- Relevant Coursework: Machine Learning, Design and Analysis of Algorithms, Database Systems, Database System Implementation, Systems and Networks, Computer Networking, Linear Algebra

EXPERIENCE

Advanced Micro Devices (AMD)

Incoming - Fall 2026

Incoming Software Engineer Intern, Ryzen AI Team

San Jose, CA

eBay

May 2026 - August 2026

Software Engineer Intern

San Jose, CA

- Engineered a production Python agent using LangChain and the Claude Agent SDK to automate telemetry analysis, evidence collection, and root-cause diagnosis, accelerating incident resolution 3x.
- Architected and deployed a Java and Go reliability platform that unified investigation state and service-health signals across 3 production systems for incident triage.
- Designed GraphQL APIs spanning cloud infrastructure, observability, and ticketing systems, providing internal applications with consistent access to operational data.

Samsung Electronics America

June 2025 - August 2025

Software Engineer Intern

Mountain View, CA

- Migrated continuous integration workloads from static Amazon EC2 build agents to ephemeral Kubernetes pods on Amazon EKS, increasing deployment throughput by 50%.
- Modeled 60-day production telemetry windows in Python using Prophet and ARIMA, validating anomaly-detection signals and enabling 10x faster incident detection.
- Created a React and TypeScript application integrating product identifiers, catalog metadata, and eCommerce service metrics, reducing recurring manual SQL investigation by approximately 80%.

Fung Group at Georgia Tech Computational Science and Engineering

August 2024 - May 2025

Lead AI/ML Undergraduate Researcher

Atlanta, GA

- Fine-tuned LLaMA model in PyTorch to generate CIF crystal structures conditioned on target band gaps and formation energies.
- Generated and evaluated 900 candidate structures across 9 band-gap targets from 1.1-1.9 eV, reducing mean absolute error to 0.21-0.30 eV through oversampling and additional fine-tuning.
- Quantified model behavior across 6-12 epochs using 7 validity, EMD, and coverage metrics, measuring 12% more successful generations, 26% higher recall, and 41% lower density EMD.

Georgia Institute of Technology - College of Computing

August 2025 - Present

Teaching Assistant

Atlanta, GA

- Taught object-oriented programming, web development, and API design in a Python course serving 600+ students

PROJECTS

Course Critique | Python, Next.js, TypeScript, AWS, Firebase, SQL

- Launched and maintained a production course analytics platform serving 20,000+ Georgia Tech students per semester, translating user needs into instructor search, professor comparisons, GPA analytics, and profile features.
- Automated Python ETL and entity-reconciliation pipelines processing 100,000+ records per semester, eliminating 90% of recurring manual processing while improving data consistency.

Distributed URL Shortener | Python, Flask, SQLite, Docker, nginx

- Developed a distributed URL-shortening REST API with Python, Flask, and SQLite across 2 Dockerized backend nodes behind NGINX.
- Stress-tested the API across 400-request runs and 4-16 concurrent clients, measuring 83% higher read throughput, 47% lower p95 latency, and zero errors.

Fake eCommerce Review Detection Tool | Python, Scikit-learn, PyTorch, Kaggle

- Derived 200+ textual and metadata features from 40,000 labeled reviews for statistical fraud classification.
- Compared Logistic Regression and SVM classifiers using feature scaling, hyperparameter optimization, and held-out validation, achieving 88% and 90% accuracy, respectively.

SKILLS AND HONORS

Languages: Python, C++, Java, Go, C, SQL, TypeScript, JavaScript

Infrastructure & Tools: AWS (EC2, EKS, ECR, S3), Docker, Kubernetes, Terraform, Git, Linux, Jenkins, GitHub Actions

Frameworks & Technologies: React, Next.js, Node.js/Express, Spring Boot, GraphQL, REST APIs, PostgreSQL, MySQL, SQLite, PyTorch, scikit-learn, LangChain, Claude Agent SDK

Honors: USACO Gold Division (2022), AIME Qualifier (2021, 2022)